incorporating more company-specific information into the financial markets.[2948] In addition, the Inline XBRL requirement for the climate-related disclosures will further limit agency problems, as requirements for financial statement tagging have been observed to facilitate external monitoring of registrants through the aforementioned reduction of information processing costs.[2949]

Investors with access to XBRL analysis software may directly benefit from the availability of the climate-related disclosures in Inline XBRL, whereas other investors may indirectly benefit from the processing of Inline XBRL disclosures by asset managers and by information intermediaries such as financial analysts.[2950] In that regard, XBRL requirements for

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[2948] *See, e.g.*, Y. Cong, J. Hao & L. Zou, *The Impact of XBRL Reporting on Market Efficiency*, 28 J. Info. Sys. 181 (2014) (finding support for the hypothesis that "XBRL reporting facilitates the generation and infusion of idiosyncratic information into the market and thus improves market efficiency"); Y. Huang, J.T. Parwada, Y.G. Shan & J. Yang, *Insider Profitability and Public Information: Evidence From the XBRL Mandate* (working paper, 2019) (finding XBRL adoption levels the informational playing field between insiders and non-insiders); J. Efendi, J.D. Park & C. Subramaniam, *Does the XBRL Reporting Format Provide Incremental Information Value? A Study Using XBRL Disclosures During the Voluntary Filing Program*, 52 Abacus 259 (2016) (finding XBRL filings have larger relative informational value than HTML filings); J. Birt, K. Muthusamy & P. Bir, *XBRL and the Qualitative Characteristics of Useful Financial Information*, 30 Acct. Res. J. 107 (2017) (finding "financial information presented with XBRL tagging is significantly more relevant, understandable and comparable to non-professional investors"); S.F. Cahan, S. Chang, W.Z. Siqueira & K. Tam, *The Roles of XBRL and Processed XBRL in 10-K Readability*, J. Bus. Fin. Acct. (2021) (finding Form 10-K file size reduces readability before XBRL's adoption since 2012, but increases readability after XBRL adoption, indicating "more XBRL data improves users' understanding of the financial statements").

[2949] *See, e.g.,* P.A. Griffin, H.A. Hong, J.B. Kim & J.H. Lim, *The SEC's XBRL Mandate and Credit Risk: Evidence on a Link Between Credit Default Swap Pricing and XBRL Disclosure*, 2014 American Accounting Association Annual Meeting (2014) (attributing the negative association between XBRL information and credit default swap spreads to "(i) a reduction in firm default risk from better outside monitoring and (ii) an increase in the quality of information about firm default risk from lower information cost"); J.Z. Chen, H.A. Hong, J.B. Kim & J.W. Ryou, *Information Processing Costs and Corporate Tax Avoidance: Evidence from the SEC's XBRL Mandate*, 40 J. Acct. Pub. Pol. (2021) (finding XBRL reporting decreases likelihood of company tax avoidance, because "XBRL reporting reduces the cost of IRS monitoring in terms of information processing, which dampens managerial incentives to engage in tax avoidance behavior").

[2950] Additional information intermediaries that have used XBRL disclosures may include financial media, data aggregators and academic researchers. *See, e.g.*, Nina Trentmann, *Companies Adjust Earnings for Covid-*

public company financial statement disclosures have been observed to increase the number of companies followed by analysts, decrease analyst forecast dispersion, and, in some cases, improve analyst forecast accuracy.[2951] Should similar impacts on the analysts' informational environment arise from climate-related disclosure tagging requirements, this will likely benefit retail investors, who have generally been observed to rely on analysts' interpretation of financial disclosures rather than directly analyzing those disclosures themselves.[2952]

With respect to the Inline XBRL tagging requirements, various preparation solutions have been developed and used by operating companies to fulfill their structuring requirements, and some evidence suggests that, for smaller companies, XBRL compliance costs have decreased

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*19 Costs, But Are They Still a One-Time Expense?*, The Wall Street Journal (2020), available at https://www.wsj.com/articles/companies-adjust-earnings-for-covid-19-costs-but-are-they-still-a-one-time-expense-11600939813 (retrieved from Factiva database) (citing XBRL research software provider Calcbench as data source); *Bloomberg Lists BSE XBRL Data*, XBRL (2018), available at https://www.xbrl.org/news/bloomberg-lists-bse-xbrl-data/; R. Hoitash & U. Hoitash, *supra* note 2877. *See 2019 Pension Review First Take: Flat to Down*, Goldman Sachs Asset Management (2020) (an example of asset manager use of XBRL data), available at https://www.gsam.com/content/dam/gsam/pdfs/common/en/public/articles/2020/2019_Pension_First_Take.pdf (citing XBRL research software provider Idaciti as a data source).

2951 See, e.g., A.J. Felo, J.W. Kim & J. Lim, Can XBRL Detailed Tagging of Footnotes Improve Financial Analy'ts' Information Environment?, 28 Int'l J. Acct. Info. Sys. 45 (2018); Y. Huang, Y.G. Shan & J.W. Yang, Information Processing Costs and Stock Price Informativeness: Evidence from the XBRL Mandate, 46 Aust. J. Mgmt. 110 (2020) (finding "a significant increase of analyst forecast accuracy post-XBRL"); M. Kirk, J. Vincent & D. Williams, From Print to Practice: XBRL Extension Use and Analyst Forecast Properties (working paper, 2016) (finding "the general trend in forecast accuracy post-XBRL adoption is positive"); C. Liu, T. Wang & L.J. Yao, XBRL's Impact on Analyst Forecast Behavior: An Empirical Study, 33 J. Acct. Pub. Pol. 69 (2014) (finding "mandatory XBRL adoption has led to a significant improvement in both the quantity and quality of information, as measured by analyst following and forecast accuracy"). But see S.L. Lambert, K. Krieger & N. Mauck, Analysts' Forecasts Timeliness and Accuracy Post-XBRL, 27 Int'l. J. Acct. Info. Mgmt. 151 (2019) (finding significant increases in frequency and speed of analyst forecast announcements, but no significant increase in analyst forecast accuracy post-XBRL).

2952 *See, e.g.*, A. Lawrence, J. Ryans & E. Sun, *Investor Demand for Sell-Side Research*, 92 Acct. Rev. 123 (2017) (finding the "average retail investor appears to rely on analysts to interpret financial reporting information rather than read the actual filing"); D. Bradley, J. Clarke, S. Lee & C. Ornthanalai, *Are Analyts' Recommendations Informative? Intraday Evidence on the Impact of Time Stamp Delays*, 69 J. Fin. 645 (2014) (concluding "analyst recommendation revisions are the most important and influential information disclosure channel examined").

over time.[2953] One commenter, in opposing the proposed Inline XBRL requirements, stated that the requirements would increase costs for registrants.[2954] While we acknowledge that costs for registrants will increase as a result of the tagging requirements, this increase should be mitigated by the fact that filers subject to the final rules are already subject to Inline XBRL requirements for other disclosures in Commission filings, including financial statement disclosures and disclosures outside the financial statements.[2955] As such, the final rules do not impose Inline XBRL compliance requirements on filers that would otherwise not be subject to such requirements, and filers may be able to leverage existing Inline XBRL preparation processes and/or expertise in complying with the climate-related disclosure tagging requirements.

Many commenters agreed that the proposed structuring requirement would enable more efficient data processing and more informed investment decisions.[2956] One commenter noted

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[2953] An AICPA survey of 1,032 reporting companies with $75 million or less in market capitalization in 2018 found an average cost of $5,476 per year, a median cost of $2,500 per year, and a maximum cost of $51,500 per year for fully outsourced XBRL creation and filing, representing a 45% decline in average cost and a 69% decline in median cost since 2014. *See* AICPA, XBRL Costs for Small Reporting Companies Have Declined 45%Since 2014 (2018), available at https://us.aicpa.org/content/dam/aicpa/interestareas/frc/accountingfinancialreporting/xbrl/downloadabledocuments/xbrl-costs-for-small-companies.pdf; *see also* letter from Nasdaq; Request for Comment on Earnings Releases and Quarterly Reports, Release No. 33–10588 (Dec. 18, 2018) [83 FR 65601 (Dec. 21, 2018)] (stating that a 2018 Nasdaq survey of 151 listed registrants found an average XBRL compliance cost of $20,000 per quarter, a median XBRL compliance cost of $7,500 per quarter, and a maximum, XBRL compliance cost of $350,000 per quarter in XBRL costs).

[2954] *See* letter from Alliance Resource.

[2955] *See* 17 CFR 229.601(b)(101); 17 CFR 232.405; *see also* 17 CFR 229.601(b)(104); 17 CFR 232.406 for requirements related to tagging cover page disclosures in Inline XBRL. Beginning in July 2024, filers of most fee-bearing forms will also be required to structure filing fee information in Inline XBRL. The Commission will provide an optional web tool that will allow filers to provide those tagged disclosures without the use of Inline XBRL compliance services or software*; see* 17 CFR 229.601(b)(107); 17 CFR 232.408; Filing Fee Disclosure and Payment Methods Modernization, Release No. 33-10997 (Oct. 13, 2021) [86 FR 70166 (Dec. 9, 2021)].

[2956] *See* letters from Crowe LLP; Institute of Internal Auditors; Data Foundation; Arcadia Power, Climate & Company; MovingWorlds; Rho Impact; Trakref; Bloomberg; London Stock Exchange Group; Morningstar; MSCI; AIMCo *et al.*; CalPERS; Can. Coalition GG; Church Investment Group; CII; PRI; SCERS; Treehouse Invest.; Research Affiliates; Cedar Street Asset Management; Ceres; Corbel Capital Partners; Decatur Capital Management; Nordea Asset Management; Ethic; First Eagle; Impact Capital Managers; ICI; ICSWG; Liontrust; Nipun Capital; and Prime Buchholz.

that tagging the new disclosures in Inline XBRL would, by allowing the disclosed information to be more readily incorporated into investors' analyses, promote the efficiency of the U.S. capital markets.[2957] Another commenter stated that tagging the new disclosures would offer significant benefits to both institutional and retail investors.[2958] A different commenter indicated that the tagging requirement should enable investors to compare the adequacy of risk analysis and mitigation planning among registrants in the same economic sector.[2959]

One commenter questioned the benefits of requiring the new disclosures to be structured by asserting that investors and market participants who need to extract and analyze the disclosures required under subpart 1500 of Regulation S-K can perform the same search manually by using the appropriate Item reference as is done for current searches.[2960] However, the availability of such disclosures in a machine-readable form will allow for search and retrieval of disclosures on an automated, large-scale basis, greatly increasing the efficiency of information acquisition as compared to manual searches through unstructured formats.[2961]

Other commenters expressed concern that the potential for excessive use of extensions (i.e., custom tags) would detract from the aforementioned benefits of structured data.[2962] We agree that the inappropriate use of custom tags hinders the benefits of tagging. However, we do not believe the final rules will result in an excessive use of custom tags, because filers will be

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2957 *See* letter from Climate Advisers.

2958 *See* letter from CFA.

2959 *See* letter from IATP.

2960 *See* letter from AFPM.

2961 *See, e.g.*, Joung W. Kim & Jee Hae Lim, *The Impact of XBRL-tagged Financial Notes on Information Environment*, The 2015 Annual Summer/International Conference-Korean Accounting Association (2015) (finding block and detail tagging of financial statement footnotes in XBRL filings improve the readability of 10-K filings and the explanatory power of certain accounting figures like net income and book value of equity on stock price).

2962 *See* letters from BHP; Morningstar; and Ethic.

prohibited from using custom tags unless there is no suitable standard tag for their disclosure in the related climate taxonomy, which the Commission will publish before the tagging compliance date.[2963] The climate taxonomy will contain standard tags that cover each new disclosure provision, so we do not expect custom tagging for climate disclosures will be excessive. Also, as discussed above, the one-year transition period for tagging requirements will enable the climate taxonomy development process to leverage samples of climate disclosures in Commission filings to further build out the list of standard tags and adapt to common disclosure practices. This should further reduce the likelihood of excessive custom tags and thus improve data quality.[2964] Such improvement in data quality will come at the cost of data users having one less year of tagged climate disclosures, making the climate disclosures filed during that year more difficult to analyze efficiently.

### 3. Quantifiable Direct Costs on Registrants

In this section, we attempt to quantify the direct costs of compliance for registrants that will be impacted by the final rules.[2965] These costs could be incurred internally (*e.g.,* through employee hours or hiring additional staff) or externally (*e.g.,* via third-party service providers, such as auditors or consultants).

Our estimates are informed, in part, by feedback we received from public comment letters. As discussed below, however, commenters offered a wide range of cost estimates, suggesting that there is significant heterogeneity when it comes to expected compliance costs

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[2963] *See* 17 CFR 232.405(c)(1)(iii)(B). Studies have found informational benefits resulting from the proper use of custom tags. *See, e.g.*, Joseph Johnston, *Extended XBRL Tags and Financial Analysts' Forecast Error and Dispersion*, 34 J. of Info. Sys. 105 (Sept. 2020) (finding custom tags to be "robustly negatively related to analysts' forecast error and dispersion").

[2964] *See supra* section II.O.

[2965] *See supra* section IV.C.1.b.

among registrants, and such estimates may not provide a representative view of the costs of compliance for all affected registrants.

The cost estimates submitted by commenters varied considerably depending on a given company's size, industry, complexity of operations, and other characteristics. This variability adds to the challenges in estimating compliance costs. Additionally, many commenters provided aggregate cost estimates that did not include certain elements required by the final rules, or included other elements that are not required in the final rules,[2966] without providing a breakdown of the component costs. Without a breakdown of component costs, it is difficult to use these cost estimates to quantify the direct cost of the final rules. Furthermore, changes from the proposal, often in response to commenter concerns about costs, will result in corresponding differences in the anticipated cost of the final rules as compared to the proposal.[2967] Nonetheless, we have endeavored below to factor these comments into our analysis to determine registrants' approximate cost of compliance with the final rules.

**a. Comments and data on direct cost estimates of the proposed rules**

In the Proposing Release, the Commission requested comment on all aspects of its economic analysis, including the potential costs and benefits of the proposed rules and alternatives, and whether the proposed rules, if adopted, will promote efficiency, competition, and capital formation or have an impact on investor protection.[2968] The Commission specifically

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[2966] Commenters' estimates that include the cost of voluntarily undertaking a specific activity (e.g., the costs of setting targets or goals, formulating transition plans, or conducting scenario analysis) may not be indicative of the compliance costs of the final rules since the final rules do not necessarily require the undertaking of such activities, but rather require only the attendant disclosures in certain cases.

[2967] For example, as compared to the proposed rules, the final rules include a number of changes intended to reduce the burden of the Regulation S-X disclosure requirements and do not require Scope 3 emissions reporting.

[2968] *See* Proposing Release, section IV.G.

requested empirical data, estimation methodologies, and other factual support for commenters' views, in particular, on costs and benefits estimates.[2969]

We received many comments asserting that the direct costs imposed by the proposed rules would be much greater than the Commission estimated.[2970] Many letters from individual companies and industry groups provided quantitative estimates of the cost to comply with the proposed rules that were considerably higher than the estimates included in the Proposing Release.[2971] One commenter conducted a survey of 263 public companies between April and June 2022.[2972] Seventy-nine percent of non-SRC respondents in this survey asserted that the Commission under-estimated the costs of compliance with the proposed rules. Seventy-three percent of survey participants responded that their compliance costs under the proposed rules would exceed the Commission's estimates in the Proposing Release, with 41 percent of respondents stating that the compliance costs would exceed $1 million on an ongoing basis.[2973] Another commenter, a biotechnology trade association, surveyed its members and found that 56 percent of respondents expected that the proposed rules would be more expensive than the Commission's estimates, with 40 percent indicating it would cost between $0.5 and $1.0 million.[2974] Additionally, a survey of corporate executives indicated that 61 percent of

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2969 *See id*.

2970 *See, e.g.,* letters from Soc. Corp. Gov. (June 17, 2022); Chamber; Business Roundtable; S.P. Kothari, *et al.*; Biotechnology Innovation Organization; Committee on Corporate Reporting; American Automotive Leasing Association (AALA); America Car Rental Association; Truck Renting and Leasing Association (TRALA); AEPC. Some commenters also critiqued our PRA analysis, asserting that it used the wrong cost of labor and did not include the costs to non-registrants. *See* letter from the Heritage Foundation.

2971 *See, e.g.*, letters from Soc. Corp. Gov. (June 17, 2022); RILA; NRF; ConocoPhillips; API; PPL Corporation; Nutrien; and Chamber.

2972 *See* letter from Nasdaq.

2973 *Id.* Twelve percent of the participants in the survey were SRCs.

2974 *See* letter from Biotechnology Innovation Organization.

respondents expect that the proposed rules would impose $750,000 or more in first year compliance costs.[2975] Some commenters specifically identified the GHG emissions reporting and Regulation S-X provisions of the proposed rules as likely to impose large cost burdens on both registrants and potentially on non-registrants.[2976]

To help assess the direct costs of the final rules, we conducted a detailed review of compliance cost estimates from commenters and other public sources. The nature of the cost information ranged from survey results, estimates directly from identifiable companies, estimates of anonymous companies, and general estimates, either based on industry experience, fees for related services, or derived as part of similar rulemaking processes in other jurisdictions. We describe below the cost estimates provided in these letters and other sources.

One commenter provided cost information from seven large-cap companies in various industries on their current voluntary climate-reporting practices, which vary in their degrees of alignment with the proposed or final rules.[2977] The responses varied considerably regarding the reporting activities, disclosure elements, and costs. The number of staff required to produce the voluntary disclosures ranged from two to 20 full-time equivalents ("FTEs"). Reported employee hours for climate reporting (including TCFD reporting) ranged from 7,500 to 10,000 hours

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[2975] PwC, *Change in the Climate: How US business leaders are preparing for the SEC's climate disclosure rule* (2023), available at https://www.pwc.com/us/en/services/esg/library/sec-climate-disclosure-survey.html (discussing survey, conducted between Dec. 2022 and Jan. 2023, that solicits the views of 300 executives at U.S.-based public companies with at least $500 million in annual revenue with respect to the proposed rules).

[2976] *See, e.g.*, letter from Chamber. Concerns about burdens for non-registrants were mostly focused on the proposed rules' Scope 3 GHG emissions disclosure requirements. The final rules do not require disclosures of Scope 3 emissions.

[2977] *See* letter from Soc. Corp. Gov (June 11, 2022), referencing a comment it submitted in response to Acting Chair Allison Herren Lee's request for public input on climate disclosures. *See* Acting Chair Allison Herren Lee Public Statement, *Public Input Welcomed on Climate Change Disclosures*, available at https://www.sec.gov/news/public-statement/lee-climate-change-disclosures. Comment letters in response to this request are available at https://www.sec.gov/comments/climate-disclosure/cll12.htm.

annually. One company reported spending 9 months to prepare its TCFD report and 4 months responding to the CDP questionnaire. Commonly cited external advisory services include environmental engineering consultants; emissions, climate science, and modeling consultants; outside counsel; and sustainability or sustainability reporting consultants, with costs ranging from $50,000 to $1.35 million annually. Third-party assurance costs ranged from $10,000 to $600,000. One company reported that it incurred initial costs of approximately $1.3 million to establish a baseline for SASB and TCFD reporting, while another company estimated that new or enhanced systems, controls, audit, and other costs associated with any additional disclosure requirements would be over $1 million.

The same commenter submitted another letter presenting detailed annual cost estimates from 13 companies (11 large-cap, 1 mid-cap, and 1 small-cap).[2978] Similar to their first comment, the responses displayed considerable variation with respect to current disclosure scope, granularity, and reported costs. Specific estimates of initial costs to comply with the proposed rules included $5 to $10 million,[2979] $6 million (with $4 to $5 million in ongoing costs),[2980] $10 million (with "much of it recurring"),[2981] and $650,000 to $1.5 million (with $650,000 in ongoing costs).[2982] In many cases, the reported costs in this comment letter aggregated several different disclosure items and related activities without providing a cost

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2978 *See* letter from Soc. Corp. Gov (June 17, 2022). The commenter acknowledges that these companies are "not the norm. They represent a discrete subset of predominantly larger companies that have undertaken these reporting efforts voluntarily and generally reflect a much greater level of maturity in climate-related reporting than the average company."

2979 *Id. See* Company 1 (large-cap company).

2980 *Id. See* Company 2 (large-cap company). Throughout this release, "ongoing costs" refer to recurring costs on an annual basis.

2981 *Id. See* Company 3 (large-cap company).

2982 *Id. See* Company 4 (small-cap company).

breakdown. In other cases, costs were much more specific. For example, some companies reported their costs of measuring emissions. One small-cap company estimated $300,000 annually in internal staff time for its Scope 1 emissions data collection and reporting. This letter also included the aggregate ongoing costs of measuring Scope 1, Scope 2, and some Scope 3 emissions[2983] from three different companies. These companies' respective estimates are $200,000,[2984] $75,000,[2985] and 188 internal hours.[2986] Other specific cost estimates included assurance (ranging from $10,000 to $550,000, depending on the scope and level of assurance), external consultants (ranging from $55,000 to $990,000), and other activities related to sustainability reporting.

A public report presents detailed climate-related reporting cost estimates from three anonymous companies.[2987] One company, a large-cap financial institution, reported that the cost of issuing their first TCFD report was less than $100,000 and that annual ongoing costs for responding to the CDP questionnaire is likewise less than $100,000. Another company, a mid-cap waste management company, stated that the cost of producing their first TCFD and SASB report were both less than $10,000. This company reported that its total annual employee costs associated with climate disclosure are approximately $12,600. It also reported incurring annual third-party costs between $60,000 to $160,000 to "develop [its] corporate sustainability report

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2983 Disclosing "some Scope 3 emissions" generally means that the commenter discloses some—but not all—categories of Scope 3 emissions. For example, one company "discloses Scope 1 and Scope 2 and some Scope 3 (fuel and energy-related activities, business travel, and use of sold products) GHG emissions…" *See* letter from Soc. Corp. Gov (June 17, 2022).

2984 *Id. See* Company 9 (large-cap company).

2985 *Id. See* Company 11 (large-cap company).

2986 *Id. See* Company 10 (large-cap company).

2987 *See* L. Reiners & K. Torrent, *The Cost of Climate Disclosure: Three Case Studies on the Cost of Voluntary Climate-Related Disclosure*, Climate Risk Disclosure Lab (2021), available at https://econ.duke.edu/sites/econ.duke.edu/files/documents/The%20Cost%20of%20Climate%20Disclosure.pdf. This source was also reviewed as part of the proposed rules. *See* Proposing Release, section IV.C.2.a.